

4.5 Generalizing LATE

The LATE theorem applies to a stripped-down causal model in which a single dummy instrument is used to estimate the impact of a dummy treatment with no covariates. We can generalize this in three important ways: multiple instruments (e.g., a set of quarter-of-birth dummies), models with covariates (e.g., controls for year of birth), and models with variable and continuous treatment intensity (e.g., years of schooling). In all three cases, the IV estimand is a weighted average of causal effects for instrument-specific compliers. The econometric tool remains 2SLS and the interpretation remains fundamentally similar to the basic LATE result, with a few bells and whistles. 2SLS with multiple instruments produces a causal effect that averages IV estimands using the instruments one at a time; 2SLS with covariates produces an average of covariate-specific LATEs; 2SLS with variable or continuous treatment intensity produces a weighted average derivative along the length of a possibly nonlinear causal response function. These results provide a simple casual interpretation for 2SLS in most empirically relevant settings.

4.5.1 *LATE with Multiple Instruments*

The multiple-instruments extension is easy to see. This is essentially the same as a result we discussed in the grouped data context. Consider a pair of dummy instruments, z_{1i} and z_{2i} . Without loss of generality, assume these dummies are mutually exclusive (if not, then we can work with a mutually exclusive set of three dummies, $z_{1i}(1 - z_{2i})$, $z_{2i}(1 - z_{1i})$, and $z_{1i}z_{2i}$). The two dummies can be used to construct Wald estimators. Again without loss of generality, assume monotonicity is satisfied for each with a positive first stage (if not, we can recode the dummies so this is true). Both therefore estimate a version of $E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]$, though the population with $D_{1i} > D_{0i}$ differs for z_{1i} and z_{2i} .

Instead of Wald estimators, we can use z_{1i} and z_{2i} together in a 2SLS procedure. Since these two dummies and a constant

exhaust the information in the instrument set, this 2SLS procedure is the same as grouped data estimation using conditional means defined given z_{1i} and z_{2i} (whether or not the instruments are correlated). As in Angrist (1991), the resulting grouped data estimator is a linear combination of the underlying Wald estimators. In other words, it is a linear combination of the instrument-specific LATEs using the instruments one at a time (in fact, it is the efficient linear combination in a traditional homoskedastic linear constant effects model).

This argument is not quite complete, since we haven't shown that the linear combination of LATEs produced by 2SLS is also a weighted average (i.e., the weights are non-negative and sum to one). The relevant weighting formulas appear in Imbens and Angrist (1994) and Angrist and Imbens (1995). The general formulas are a little messy, so here we lay out a simple version based on the two-instrument example. The example shows that 2SLS using z_{1i} and z_{2i} together is a weighted average of IV estimates using z_{1i} and z_{2i} one at a time. Let

$$\rho_j = \frac{\text{Cov}(y_i, z_{ji})}{\text{Cov}(D_i, z_{ji})}; j = 1, 2$$

denote the two IV estimands using z_{1i} and z_{2i} .

The (population) first-stage fitted values for 2SLS are $\hat{d}_i = \pi_{11}z_{1i} + \pi_{12}z_{2i}$, where π_{11} and π_{12} are positive numbers. By virtue of the IV interpretation of 2SLS, the 2SLS estimand is

$$\begin{aligned} \rho_{2SLS} &= \frac{\text{Cov}(y_i, \hat{d}_i)}{\text{Cov}(D_i, \hat{d}_i)} = \frac{\pi_{11}\text{Cov}(y_i, z_{1i})}{\text{Cov}(D_i, \hat{d}_i)} + \frac{\pi_{12}\text{Cov}(y_i, z_{2i})}{\text{Cov}(D_i, \hat{d}_i)} \\ &= \left[\frac{\pi_{11}\text{Cov}(D_i, z_{1i})}{\text{Cov}(D_i, \hat{d}_i)} \right] \left[\frac{\text{Cov}(y_i, z_{1i})}{\text{Cov}(D_i, z_{1i})} \right] \\ &\quad + \left[\frac{\pi_{12}\text{Cov}(D_i, z_{2i})}{\text{Cov}(D_i, \hat{d}_i)} \right] \left[\frac{\text{Cov}(y_i, z_{2i})}{\text{Cov}(D_i, z_{2i})} \right] \\ &= \psi\rho_1 + (1 - \psi)\rho_2, \end{aligned}$$

where ψ_1 is LATE using z_{1i} and ρ_2 is LATE using z_{2i} , and

$$\psi = \frac{\pi_{11}\text{Cov}(D_i, z_{1i})}{\pi_{11}\text{Cov}(D_i, z_{1i}) + \pi_{12}\text{Cov}(D_i, z_{2i})}$$

is a number between zero and one that depends on the relative strength of each instrument in the first stage. Thus, we have shown that 2SLS is a weighted average of causal effects for instrument-specific compliant subpopulations. Suppose, for example, that z_{1i} denotes twin births and z_{2i} indicates (nontwin) same-sex sibships in families with two or more children, both instruments for family size, as in Angrist and Evans (1998). A multiple second birth increases the likelihood of having a third child by about .6, while a same-sex sibling pair increases the likelihood of a third birth by about .07. When these two instruments are used together, the resulting 2SLS estimates are a weighted average of the Wald estimates produced by using the instruments one at a time.²⁸

4.5.2 *Covariates in the Heterogeneous Effects Model*

You might be wondering where the covariates have gone. After all, covariates played a starring role in our earlier discussion of regression and matching. Yet the LATE theorem does not involve covariates. This stems from the fact that when we see instrumental variables as a type of (natural or man-made) randomized trial, covariates take a back seat. If the instrument is randomly assigned, it is likely to be independent of covariates. Not all instruments have this property, however. As with covariates in the regression models in the previous chapter, the main reason why covariates are included in causal analyses using instrumental variables is that the conditional independence and exclusion restrictions underlying IV estimation may be more likely to be valid after conditioning on covariates. Even randomly assigned instruments, like draft eligibility status, may be valid only after conditioning on covariates. In the case of draft eligibility, older cohorts were more likely to be draft eligible because the draft eligibility cutoffs were higher. Because there are year-of-birth (or age) differences in earnings,

²⁸Using twins instruments alone, the IV estimate of the effect of a third child on female labor force participation is $-.084$. The corresponding same-sex estimate is $-.138$. Using both instruments produces a 2SLS estimate of $-.098$. The 2SLS weight in this case is .74 for twins, .26 for same sex, due to the much stronger twins first stage.

draft eligibility is a valid instrument only after conditioning on year of birth.

More formally, IV estimation with covariates may be justified by a *conditional* independence assumption

$$\{Y_{1i}, Y_{0i}, D_{1i}, D_{0i}\} \perp\!\!\!\perp Z_i | X_i \quad (4.5.1)$$

In other words, we think of the instrumental variables as being “as good as randomly assigned,” conditional on covariates X_i (here we are implicitly maintaining the exclusion restriction as well). A second reason for incorporating covariates is that conditioning on covariates may reduce some of the variability in the dependent variable. This can lead to more precise 2SLS estimates.

A benchmark constant effects model with covariates imposes functional form restrictions as follows:

$$E[Y_{0i}|X_i] = X_i' \alpha^* \text{ for a } \kappa \times 1 \text{ vector of coefficients, } \alpha^*;$$

$$Y_{1i} - Y_{0i} = \rho.$$

In combination with (4.5.1), this motivates 2SLS estimation of an equation like (4.1.6), as discussed in section 4.1.

A straightforward generalization of the constant effects model allows

$$Y_{1i} - Y_{0i} = \rho(X_i),$$

where $\rho(X_i)$ is a deterministic function of X_i . This model can be estimated by adding interactions between Z_i and X_i to the first stage and (the same) interactions between D_i and X_i to the second stage. There are now multiple endogenous variables and hence multiple first-stage equations. These can be written

$$D_i = X_i' \pi_{00} + \pi_{01} Z_i + Z_i X_i' \pi_{02} + \xi_{0i} \quad (4.5.2a)$$

$$D_i X_i = X_i' \pi_{10} + \pi_{11} Z_i + Z_i X_i' \pi_{12} + \xi_{1i}. \quad (4.5.2b)$$

Although (4.5.2b) is written as if $D_i X_i$ is a scalar, there should be a first stage like this for each element of $D_i X_i$. The second-stage equation in this case is

$$Y_i = \alpha' X_i + \rho_0 D_i + D_i X_i' \rho_1 + \eta_i,$$

so $\rho(X_i) = \rho_0 + \rho_1'X_i$. Alternatively, a nonparametric version of $\rho(X_i)$ can be estimated by 2SLS in subsamples stratified on X_i .

The heterogeneous effects model underlying the LATE theorem allows for identification based on conditional independence as in (4.5.1), though the interpretation is a little more complicated than for LATE without covariates. For each value of X_i , we define covariate-specific LATE,

$$\lambda(X_i) \equiv E[Y_{1i} - Y_{0i} | X_i, D_{1i} > D_{0i}].$$

The “saturate and weight” approach to estimation with covariates, which generates a weighted average of $\lambda(X_i)$, is spelled out in the following theorem (from Angrist and Imbens, 1995).

Theorem 4.5.1 *Saturate and Weight. Suppose the assumptions of the LATE theorem hold conditional on X_i . That is,*

(CA1, Independence) $\{Y_i(D_{1i}, 1), Y_{0i}(D_{0i}, 0), D_{1i}, D_{0i}\} \perp\!\!\!\perp Z_i | X_i$;

(CA2, Exclusion) $P[Y_i(d, 0) = Y_i(d, 1) | X_i] = 1$ for $d = 0, 1$;

(CA3, First Stage) $E[D_{1i} - D_{0i} | X_i] \neq 0$.

We also assume monotonicity (A4) holds as before. Consider the 2SLS estimand based on the first-stage equation

$$D_i = \pi_X + \pi_{1X}Z_i + \xi_{1i} \quad (4.5.3)$$

and the second-stage equation

$$Y_i = \alpha_X + \rho_c D_i + \eta_i,$$

where π_X and α_X denote saturated models for covariates (a full set of dummies for all values of X_i) and π_{1X} denotes a first-stage effect of Z_i for every value of X_i . Then $\rho_c = E[\omega(X_i)\lambda(X_i)]$, where

$$\omega(X_i) = \frac{V\{E[D_i | X_i, Z_i] | X_i\}}{E[V\{E[D_i | X_i, Z_i] | X_i\}]} \quad (4.5.4)$$

and

$$V\{E[D_i | X_i, Z_i] | X_i\} = E\{E[D_i | X_i, Z_i](E[D_i | X_i, Z_i] - E[D_i | X_i]) | X_i\}.$$

This theorem says that 2SLS with a fully saturated first stage and a saturated model for covariates in the second stage produces a weighted average of covariate-specific LATEs. The weights are proportional to the average conditional variance of the population first-stage fitted value, $E[D_i|X_i, Z_i]$, at each value of X_i .²⁹ The theorem comes from the fact that the first stage coincides with $E[D_i|X_i, Z_i]$ when (4.5.3) is saturated (i.e., the first-stage regression recovers the CEF).

In practice, we may not want to work with a model with a first-stage parameter for each value of the covariates. First, there is the risk of bias, as we discuss at the end of this chapter, and second, a big pile of individually imprecise first-stage estimates is not pretty to look at. It seems reasonable to imagine that models with fewer parameters, say a restricted first stage imposing a constant π_{1X} , nevertheless approximate some kind of covariate-averaged LATE. This turns out to be true, but the argument is surprisingly indirect. The vision of 2SLS as providing a MMSE approximation to an underlying causal relation was developed by Abadie (2003).

The Abadie approach begins by defining the object of interest to be $E[Y_i|D_i, X_i, D_{1i} > D_{0i}]$, the CEF for Y_i given treatment status and covariates, for compliers. An important feature of this CEF is that when the conditions of the LATE theorem hold conditional on X_i , it has a causal interpretation. In other words, *for compliers*, treatment-control contrasts conditional on X_i are equal to conditional-on- X_i LATEs:

$$\begin{aligned} E[Y_i|D_i = 1, X_i, D_{1i} > D_{0i}] - E[Y_i|D_i = 0, X_i, D_{1i} > D_{0i}] \\ = E[Y_{1i} - Y_{0i}|X_i, D_{1i} > D_{0i}] \\ = \lambda(X_i). \end{aligned}$$

This follows immediately from the facts that $D_i = Z_i$ for compliers and, given (4.5.1), potential outcomes are independent of Z_i given X_i and $D_{1i} > D_{0i}$. The upshot is that a regression of Y_i on D_i and X_i in the complier population also has a causal interpretation. Although this regression might not

²⁹Note that the variability in $E[D_i|X_i, Z_i]$ conditional on X_i comes from Z_i . So the weighting formula gives more weight to covariate values where the instrument creates more variation in fitted values.

give us the CEF of interest (unless it is linear or the model is saturated), it will, as always, provide the MMSE approximation to it. That is, a regression of Y_i on D_i and X_i in the complier population approximates $E[Y_i|D_i, X_i, D_{1i} > D_{0i}]$, just as OLS approximates $E[Y_i|D_i, X_i]$. Alas, we do not know who the compliers are, so we cannot sample them. Nevertheless, they can be found, in the following sense.

Theorem 4.5.2 Abadie Kappa. *Suppose the assumptions of the LATE theorem hold conditional on covariates, X_i . Let $g(Y_i, D_i, X_i)$ be any measurable function of (Y_i, D_i, X_i) with finite expectation. Define*

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - P(Z_i = 1|X_i)} - \frac{(1 - D_i)Z_i}{P(Z_i = 1|X_i)}.$$

Then

$$E[g(Y_i, D_i, X_i)|D_{1i} > D_{0i}] = \frac{E[\kappa_i g(Y_i, D_i, X_i)]}{E[\kappa_i]}.$$

This can be proved by direct calculation using the fact that, given the assumptions of the LATE theorem, any expectation is a weighted average of means for always-takers, never-takers, and compliers. By monotonicity, those with $D_i(1 - Z_i) = 1$ are always-takers because they have $D_{0i} = 1$, while those with $(1 - D_i)Z_i = 1$ are never-takers because they have $D_{1i} = 0$. Hence, the compliers are the left-out group.

The Abadie theorem has a number of important implications; for example, it crops up again in the discussion of quantile treatment effects. Here, we use it to approximate $E[Y_i|D_i, X_i, D_{1i} > D_{0i}]$ by linear regression. Specifically, let α_c and β_c solve

$$\begin{aligned} (\alpha_c, \beta_c) = \arg \min_{a,b} E\{ & (E[Y_i|D_i, X_i, D_{1i} > D_{0i}] \\ & - aD_i - X_i'b)^2 | D_{1i} > D_{0i} \}. \end{aligned}$$

In other words, $\alpha_c D_i + X_i' \beta_c$ gives the MMSE approximation to $E[Y_i|D_i, X_i, D_{1i} > D_{0i}]$, or fits it exactly if it's linear. A consequence of Abadie's theorem is that this approximating

function can be obtained by solving

$$(\alpha_c, \beta_c) = \arg \min_{a,b} E\{\kappa_i(Y_i - aD_i - X_i'b)^2\}, \quad (4.5.5)$$

the kappa-weighted least squares minimand.³⁰

Abadie proposes an estimation strategy (and develops distribution theory) for a procedure that involves first-step estimation of κ_i using parametric or semiparametric models for $P(z_i = 1|X_i)$. The estimates from the first step are then plugged into the sample analog of (4.5.5) in the second step. Not surprisingly, when the only covariate is a constant, Abadie's procedure simplifies to the Wald estimator. More surprisingly, minimization of (4.5.5) produces the traditional 2SLS estimator as long as a linear model is used for $P(z_i = 1|X_i)$ in the construction of κ_i . In other words, if $P(z_i = 1|X_i) = X_i'\pi$ is used when constructing an estimate of κ_i , the Abadie estimand is 2SLS. Thus, we can conclude that whenever $P(z_i = 1|X_i)$ can be fit or closely approximated by a linear model, it makes sense to view 2SLS as an approximation to the complier causal response function, $E[Y_i|D_i, X_i, D_{0i} > D_{0i}]$. On the other hand, α_a is not, in general, the 2SLS estimand, and β_a is not, in general, the vector of covariate effects produced by 2SLS. Still, the equivalence to 2SLS for linear $P(z_i = 1|X_i)$ leads us to think that Abadie's method and 2SLS are likely to produce similar estimates in most applications.

The Angrist (2001) reanalysis of Angrist and Evans (1998) is an example where estimates based on (4.5.5) are indistinguishable from 2SLS estimates. Using twins instruments to estimate the effect of a third child on female labor supply generates a 2SLS estimate of $-.088$, while the corresponding Abadie estimate is $-.089$. Similarly, 2SLS and Abadie estimates of the effect on hours worked are identical at -3.55 . This is not a strike against Abadie's procedure. Rather, it supports

³⁰The class of approximating functions needn't be linear. Instead of $\alpha_c D_i + X_i' \beta_c$, it might make sense to use a nonlinear function such as an exponential (if the dependent variable is non-negative) or probit (if the dependent variable is zero-one). We return to this point at the end of this chapter. As noted in section 4.4.4, the kappa-weighting scheme can be used to characterize covariate distributions for compliers as well as to estimate outcome distributions.

the notion that 2SLS approximates the causal relation of interest.³¹

4.5.3 Average Causal Response with Variable Treatment Intensity★

An important difference between the causal effects of a dummy variable and those of a variable that takes on the values $\{0, 1, 2, \dots\}$ is that in the first case, there is only one causal effect for any one person, while in the latter there are many: the effect of going from 0 to 1, the effect of going from 1 to 2, and so on. The potential outcomes notation we used for schooling recognizes this. Here it is again. Let

$$Y_{si} \equiv f_i(s)$$

denote the potential (or latent) earnings that person i would receive after obtaining s years of education. Note that the function $f_i(s)$ has an “ i ” subscript on it but s does not. The function $f_i(s)$ tells us what i would earn for *any* value of schooling, s , and not just for the realized value, s_i . In other words, $f_i(s)$ answers causal “what if” questions for multinomial s_i .

Suppose that s_i takes on values in the set $\{0, 1, \dots, \bar{s}\}$. Then there are $\bar{s}$ unit causal effects, $Y_{si} - Y_{s-1,i}$. A linear causal model assumes these are the same for all s and for all i , obviously unrealistic assumptions. But we need not take these assumptions literally. Rather, 2SLS provides a computational device that generates a weighted average of unit causal effects, with a weighting function we can estimate and study, so as to learn

³¹The Abadie estimator can be computed by weighting conventional linear or nonlinear regression software. The trick is to first construct a weighting scheme with positive weights. This is accomplished by iterating expectations in (4.5.5), so that κ_i (which is negative for always-takers and never-takers) can be replaced by the always-positive average weight,

$$E[\kappa_i | X_i, D_i, Y_i] = 1 - \frac{D_i(1 - E[Z_i | X_i, D_i, Y_i])}{1 - P(Z_i = 1 | X_i)} - \frac{(1 - D_i)E[Z_i | X_i, D_i, Y_i]}{P(Z_i = 1 | X_i)}.$$

(See also the discussion in section 7.2.1.) Abadie (2003) gives formulas for standard errors and Alberto Abadie has posted software to compute them, as well as the corresponding parameter estimates. Standard errors for the Abadie estimator can also be estimated using a bootstrap.

where the action is coming from with a particular instrument. This weighting function tells us how the compliers are distributed over the range of s_i . It tells us, for example, that the returns to schooling estimated using quarter of birth or compulsory schooling laws come from shifts in the distribution of grades completed in high school. Other instruments, such as the distance instruments used by Card (1995), act elsewhere on the schooling distribution and therefore capture a different sort of return.

To flesh this out, suppose that a single binary instrument, z_i , a dummy for having been born in a state with restrictive compulsory attendance laws, is to be used to estimate the returns to schooling (as in Acemoglu and Angrist, 2000). Also, let s_{1i} denote the schooling i would get if $z_i = 1$, and let s_{0i} denote the schooling i would get if $z_i = 0$. The theorem below, from Angrist and Imbens (1995), offers an interpretation of the Wald estimand with variable treatment intensity in this case. Note that here we combine the independence and exclusion restrictions by simply stating that potential outcomes indexed by s are independent of the instruments.

Theorem 4.5.3 *Average Causal Response. Suppose*

(ACR1, *Independence and Exclusion*) $\{Y_{0i}, Y_{1i}, \dots, Y_{\bar{s}i}; s_{0i}, s_{1i}\} \perp\!\!\!\perp Z_i$;

(ACR2, *First Stage*) $E[s_{1i} - s_{0i}] \neq 0$;

(ACR3, *Monotonicity*) $s_{1i} - s_{0i} \geq 0 \forall i$, or vice versa; assume the first.

Then

$$\begin{aligned} & \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[s_i|Z_i = 1] - E[s_i|Z_i = 0]} \\ &= \sum_{s=1}^{\bar{s}} \omega_s E[Y_{si} - Y_{s-1,i} | s_{1i} \geq s > s_{0i}], \end{aligned}$$

where

$$\omega_s = \frac{P[s_{1i} \geq s > s_{0i}]}{\sum_{j=1}^{\bar{s}} P[s_{1i} \geq j > s_{0i}]}.$$

The weights ω_s are non-negative and sum to one.

The average causal response (ACR) theorem says that the Wald estimator with variable treatment intensity is a weighted average of the *unit causal response* along the length of the potentially nonlinear causal relation described by $f_i(s)$. The unit causal response, $E[Y_{si} - Y_{s-1,i} | s_{1i} \geq s > s_{0i}]$, is the average difference in potential outcomes for *compliers at point s*, that is, individuals driven by the instrument from a treatment intensity less than s to at least s . For example, the quarter-of-birth instruments used by Angrist and Krueger (1991) push some people from 11th grade to finishing 12th or higher, and others from 10th grade to finishing 11th or higher. The Wald estimator using quarter-of-birth instruments combines all these effects into a single ACR.

The size of the group of compliers at point s is $P[s_{1i} \geq s > s_{0i}]$. By monotonicity, this must be non-negative and is given by the difference in the CDF of s_i at point s . To see this, note that

$$\begin{aligned} P[s_{1i} \geq s > s_{0i}] &= P[s_{1i} \geq s] - P[s_{0i} \geq s] \\ &= P[s_{0i} < s] - P[s_{1i} < s], \end{aligned}$$

which is non-negative since monotonicity requires $s_{1i} \geq s_{0i}$. Moreover,

$$P[s_{0i} < s] - P[s_{1i} < s] = P[s_i < s | z_i = 0] - P[s_i < s | z_i = 1]$$

by independence. Finally, note that because the mean of a non-negative random variable is the sum (or integral) of one minus the CDF, we have,

$$\begin{aligned} E[s_i | z_i = 1] - E[s_i | z_i = 0] &= \sum_{j=1}^{\bar{s}} (P[s_i < j | z_i = 0] - P[s_i < j | z_i = 1]) \\ &= \sum_{j=1}^{\bar{s}} P[s_{1i} \geq j > s_{0i}] \end{aligned}$$

Thus, the ACR weighting function can be consistently estimated by comparing the CDFs of the endogenous variable (treatment intensity) with the instrument switched off and on. The weighting function is normalized by the first stage.

The ACR theorem helps us understand what we are learning from a 2SLS estimate. For example, instrumental variables derived from compulsory attendance and child labor laws capture the causal effect of increases in schooling in the 6–12 grade range, but tell us little about the effects of postsecondary schooling. This is illustrated in figure 4.5.1, taken from Acemoglu and Angrist (2000).

The figure plots differences in the probability that educational attainment is at or exceeds the grade level on the x -axis (i.e., one minus the CDF). The differences are between men exposed to different child labor laws and compulsory schooling laws in the sample of white men aged 40–49 drawn from the 1960, 1970, and 1980 censuses. The instruments are coded as the number of years of schooling required either to work (panel A) or to leave school (panel B) in the year the respondent was age 14. Men exposed to the least restrictive laws are the reference group. Each instrument (e.g., a dummy for seven years of schooling required before work is allowed) can be used to construct a Wald estimator by making comparisons with the reference group.

The top panel of figure 4.5.1 shows that men exposed to more restrictive child labor laws were one to six percentage points more likely to complete grades 8–12. The intensity of the shift depends on whether the laws required seven, eight, or nine-plus years of schooling before work was allowed. But in all cases, the CDF differences decline at lower grades, and drop off sharply after grade 12. The bottom panel shows a similar pattern for compulsory attendance laws, though the effects are a little smaller and the action here is at somewhat higher grades, consistent with the fact that compulsory attendance laws are typically binding in higher grades than child labor laws. Interestingly, the child labor and compulsory attendance instruments generate similar 2SLS estimates of about .08–.10.

Before wrapping up our discussion of LATE generalizations, it's worth noting that most of the elements covered here work in combination. For example, models with multiple instruments and variable treatment intensity generate a weighted average of the ACR for each instrument. Likewise, the saturate

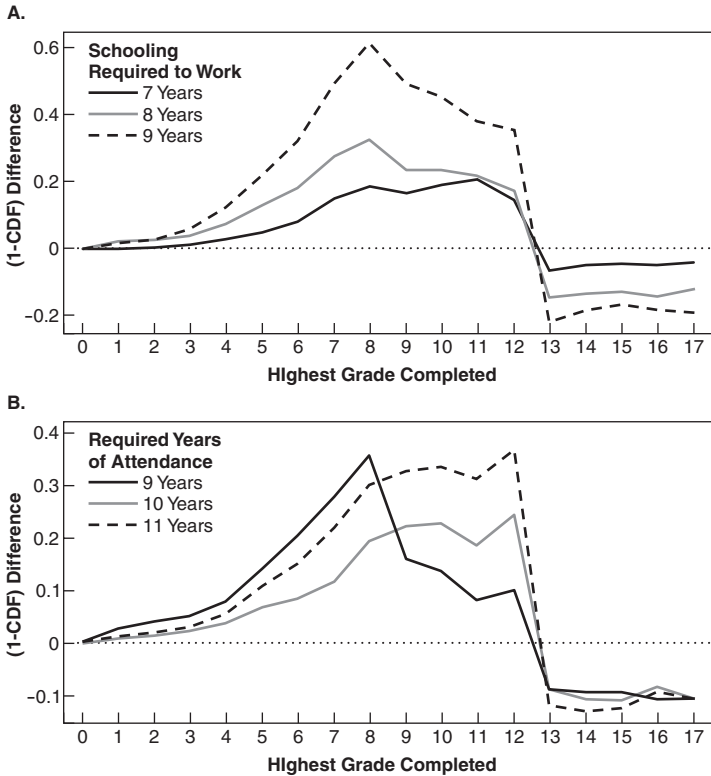


Figure 4.5.1 The effect of compulsory schooling instruments on education (from Acemoglu and Angrist 2000). The figures show the instrument-induced difference in the probability that schooling is greater than or equal to the grade level on the x -axis. The reference group is six or fewer years of required schooling in the top panel and eight or fewer years in the bottom panel. The top panel shows the CDF difference by severity of child labor laws. The bottom panel shows the CDF difference by severity of compulsory attendance laws.

and weight theorem applies to models with variable treatment intensity (though we do not yet have an extension of Abadie's kappa for models with variable treatment intensity). A final important extension covers the scenario where

the causal variable of interest is continuous and we can therefore think of the causal response function as having derivatives.

So Long, and Thanks for All the Fish

Suppose, as with the schooling problem, that counterfactuals are generated by an underlying functional relation. In this case, however, the causal variable of interest can take on any non-negative value and the functional relation is assumed to have a derivative. An example where this makes sense is a demand curve, the quantity demanded as a function of price. In particular, let $q_i(p)$ denote the quantity demanded in market i at hypothetical price p . This is a potential outcome, like $f_i(s)$, except that instead of individuals the unit of observation is a time or a location or both. For example, Angrist, Graddy, and Imbens (2000) estimate the elasticity of quantity demanded at the Fulton wholesale fish market in New York City. The slope of this demand curve is $q'_i(p)$; if quantity and price are measured in logs, this is an elasticity.

The instruments in Angrist, Graddy, and Imbens (2000) are derived from data on weather conditions off the coast of Long Island, not too far from major commercial fishing grounds. Stormy weather makes it hard to catch fish, driving up the price and reducing the quantity demanded. Angrist, Graddy, and Imbens use dummy variables such as $stormy_i$, a dummy indicating periods with high wind and waves, to estimate the demand for fish. The data consist of daily observations on wholesale purchases of whiting, a cheap fish used for fish cakes and things like that.

The Wald estimator using the $stormy_i$ instrument can be interpreted using

$$\begin{aligned} & \frac{E[q_i | stormy_i = 1] - E[q_i | stormy_i = 0]}{E[p_i | stormy_i = 1] - E[p_i | stormy_i = 0]} \\ &= \frac{\int E[q'_i(t) | p_{1i} \geq t > p_{0i}] P[p_{1i} \geq t > p_{0i}] dt}{\int P[p_{1i} \geq t > p_{0i}] dt}, \quad (4.5.6) \end{aligned}$$

where p_i is the price in market (day) i and p_{1i} and p_{0i} are potential prices indexed by *stormy* _{i} . This is a weighted average derivative with weighting function $P[p_{1i} \geq t > p_{0i}] = P[p_i < t | \text{stormy}_i = 0] - P[p_i < t | \text{stormy}_i = 1]$ at price t . In other words, IV estimation using *stormy* _{i} produces an average of the derivative $q'_i(t)$, with weight given to each possible price (indexed by t) in proportion to the instrument-induced change in the cumulative distribution function (CDF) of prices at that point. This is the same sort of averaging as in the ACR theorem except that now the underlying causal response is a derivative instead of a one-unit difference.

The continuous ACR formula, (4.5.6), comes from the fact that

$$E[q_i | \text{stormy}_i = 1] - E[q_i | \text{stormy}_i = 0] = E \left[\int_{p_{0i}}^{p_{1i}} q'_i(t) dt \right], \quad (4.5.7)$$

by the independence assumption and the fundamental theorem of calculus. Two interesting special cases fall neatly out of (4.5.7). The first is when the causal response function is linear, that is, $q_i(p) = \alpha_{0i} + \alpha_{1i}p$, for some random coefficients, α_{0i} and α_{1i} . Then, we have

$$\frac{E[q_i | \text{stormy}_i = 1] - E[q_i | \text{stormy}_i = 0]}{E[p_i | \text{stormy}_i = 1] - E[p_i | \text{stormy}_i = 0]} = \frac{E[\alpha_{1i}(p_{1i} - p_{0i})]}{E[p_{1i} - p_{0i}]}, \quad (4.5.8)$$

a weighted average of the random coefficient, α_{1i} . The weights are proportional to the price change induced by the weather in market i .

The second special case is when we can write quantity demanded as

$$q_i(p) = Q(p) + \eta_i, \quad (4.5.9)$$

where $Q(p)$ is a nonstochastic function and η_i is an additive random error. By this we mean $q'_i(p) = Q'(p)$ every day or in

every market. In this case, the average causal response function becomes

$$\int Q'(t)\omega(t)dt, \text{ where } \omega(t) = \frac{P[p_{1i} \geq t > p_{0i}]}{\int P[p_{1i} \geq r > p_{0i}]dr}$$

where r is the integrating variable in the denominator.

These special cases highlight the two types of averaging wrapped up in the ACR theorem and its continuous corollary, (4.5.6). First, there is averaging *across* markets, with weights proportional to the first-stage impact on prices in each market. Markets where prices are highly sensitive to the weather contribute the most. Second, there is averaging *along* the length of the causal response function in a given market. IV recovers the average derivative over the range of prices where the instruments shift the CDF of prices most sharply.

4.6 IV Details

4.6.1 2SLS Mistakes

2SLS estimates are easy to compute, especially since software packages like SAS and Stata will do it for you. Occasionally, however, you might be tempted to do it yourself just to see if it really works. Or you may be stranded on the planet Krikkit with all of your software licenses expired (Krikkit is encased in a slo-time envelope, so it will take you a long time to get licenses renewed). Manual 2SLS is for just such emergencies. In the manual 2SLS procedure, you estimate the first stage yourself (which in any case you should be looking at) and plug the fitted values into the second-stage equation, which is then estimated by OLS. Returning to the system at the beginning of this chapter, the first and second stages are

$$\begin{aligned} s_i &= X_i' \pi_{10} + \pi_{11}' Z_i + \xi_{1i} \\ y_i &= \alpha' X_i + \rho \hat{s}_i + [\eta_i + \rho(s_i - \hat{s}_i)], \end{aligned}$$

where X_i is a set of covariates, Z_i is a set of excluded instruments, and the first-stage fitted values are $\hat{s}_i = X_i' \hat{\pi}_{10} + \hat{\pi}_{11}' Z_i$.

Manual 2SLS takes some of the mystery out of canned 2SLS and may be useful in a software crisis, but it opens the door to mistakes. For one thing, as we discussed earlier, the OLS standard errors from the manual second stage will not be correct (the OLS residual variance is the variance of $\eta_i + \rho(s_i - \hat{s}_i)$, while for proper 2SLS standard errors you want the variance of η_i only). There are more subtle risks as well.

Covariate Ambivalence

Suppose the covariate vector contains two sorts of variables, some (say, X_{0i}) that you are comfortable with, and others (say, X_{1i}) about which you are ambivalent. Griliches and Mason (1972) faced this scenario when constructing 2SLS estimates of a wage equation that treats AFQT scores (an ability test used by the armed forces) as an endogenous variable to be instrumented. The instruments for AFQT are early schooling (completed before military service), race, and family background variables. They estimated a system that can be described like this:

$$\begin{aligned}s_i &= X'_{0i}\pi_{10} + \pi'_{11}Z_i + \xi_{1i} \\ y_i &= \alpha'_0 X_{0i} + \alpha'_1 X_{1i} + \rho \hat{s}_i + [\eta_i + \rho(s_i - \hat{s}_i)].\end{aligned}$$

This looks a lot like manual 2SLS.

A closer look, however, reveals an important difference between the equations above and the usual 2SLS procedure: the covariates in the first and second stages are not the same. For example, Griliches and Mason included age in the second stage but not in the first, a fact noted by Cardell and Hopkins (1977) in a comment on their paper. This was a mistake. Griliches and Mason's second-stage estimates are not the same as 2SLS. What's worse, they are inconsistent where 2SLS might have been fine. To see why, note that the first-stage residual, $s_i - \hat{s}_i$, is uncorrelated with X_{0i} by construction, since OLS residuals are always uncorrelated with included regressors. But because X_{1i} is not included in the first stage it is likely to be correlated with the first-stage residuals (e.g., age is probably correlated with the AFQT residual from the Griliches and Mason (1972) first stage). The inconsistency from this

correlation spills over to all coefficients in the second stage. The moral of the story: Put the same exogenous covariates in your first and second stage. If a covariate is good enough for the second stage, it's good enough for the first.

Forbidden Regressions

Forbidden regressions were forbidden by MIT professor Jerry Hausman in 1975, and while they occasionally resurface in an undersupervised thesis, they are still technically off-limits. A forbidden regression crops up when researchers apply 2SLS reasoning directly to nonlinear models. A common scenario is a dummy endogenous variable. Suppose, for example, that the causal model of interest is

$$y_i = \alpha'X_i + \rho D_i + \eta_i, \quad (4.6.1)$$

where D_i is a dummy variable for veteran status. The usual 2SLS first stage is

$$D_i = \pi'_{10}X_i + \pi'_{11}Z_i + \xi_{1i}, \quad (4.6.2)$$

a linear regression of D_i on covariates and a vector of instruments, Z_i .

Because D_i is a dummy variable, the CEF associated with this first stage, $E[D_i|X_i, Z_i]$, is probably nonlinear. So the usual OLS first stage is an approximation to the underlying nonlinear CEF. We might, therefore, use a nonlinear first stage in an attempt to come closer to the CEF. Suppose that we use probit to model $E[D_i|X_i, Z_i]$. The probit first stage is $\Phi[\pi'_{p0}X_i + \pi'_{p1}Z_i]$, where π_{p0} and π_{p1} are probit coefficients and the fitted values are $\hat{D}_{pi} = \Phi[\hat{\pi}'_{p0}X_i + \hat{\pi}'_{p1}Z_i]$. The forbidden regression in this case is the second-stage equation created by substituting $\hat{D}_{pi}$ for D_i :

$$y_i = \alpha'X_i + \rho \hat{D}_{pi} + [\eta_i + \rho(D_i - \hat{D}_{pi})]. \quad (4.6.3)$$

The problem with (4.6.3) is that only OLS estimation of (4.6.2) is guaranteed to produce first-stage residuals that are uncorrelated with fitted values and covariates. If $E[D_i|X_i, Z_i] = \Phi[X_i'\pi_{p0} + \pi'_{p1}Z_i]$, then residuals from the nonlinear model

will be asymptotically uncorrelated with X_i and $\hat{D}_{pi}$, but who is to say that the first-stage CEF is really probit? In contrast, with garden variety 2SLS, we do not need to worry about whether the first-stage CEF is really linear.³²

A simple alternative to the forbidden second step, (4.6.3), avoids problems due to an incorrect nonlinear first stage. Instead of plugging in nonlinear fitted values, we can use the nonlinear fitted values as instruments. In other words, use $\hat{D}_{pi}$ as an instrument for D_i in (4.6.1) in a conventional 2SLS procedure (as always, the exogenous covariates, X_i , should also be in the instrument list). Use of fitted values as instruments is the same as plugging in fitted values when the first stage is estimated by OLS, but not in general. Using nonlinear fits as instruments has the further advantage that, if the nonlinear model gives a better approximation to the first-stage CEF than the linear model, the resulting 2SLS estimates will be more efficient than those using a linear first stage (Newey, 1990).

But here, too, there is a drawback. The procedure using nonlinear fitted values as instruments implicitly uses nonlinearities in the first stage as a source of identifying information. To see this, suppose the causal model of interest includes the vector of instruments, Z_i :

$$Y_i = \alpha'X_i + \gamma'Z_i + \rho D_i + \eta_i. \quad (4.6.4)$$

Now, with the first stage given by (4.6.2), the model is unidentified, and conventional 2SLS estimates of (4.6.4) don't exist. In fact, 4.6.4 violates the exclusion restriction. But 2SLS estimates using X_i , Z_i , and $\hat{D}_{pi}$ as instruments do exist, because $\hat{D}_{pi}$ is a nonlinear function of X_i and Z_i that is excluded from the second stage. Should you use this nonlinearity as a source of identifying information? We usually prefer to avoid this sort of back-door identification since it's not clear what the underlying experiment really is.

³²The insight that consistency of 2SLS estimates in a traditional SEM does not depend on correct specification of the first-stage CEF goes back to Kelejian (1971). Use of a nonlinear plug-in first stage may not do too much damage in practice—a probit first stage can be pretty close to linear—but why take a chance when you don't have to?

As a rule, naively plugging in first-stage fitted values in nonlinear models is a bad idea. This includes models with a nonlinear second stage as well as those where the CEF for the first stage is nonlinear. Suppose, for example, that you believe the causal relation between schooling and earnings is approximately quadratic but otherwise homogeneous (as in Card's (1995) structural model). In other words, the model of interest is

$$y_i = \alpha'X_i + \rho_1 s_i + \rho_2 s_i^2 + \eta_i. \quad (4.6.5)$$

Given two instruments, it's easy enough to estimate (4.6.5), treating both s_i and s_i^2 as endogenous. In this case, there are two first-stage equations, one for s_i and one for s_i^2 . Although you need at least two instruments for this to work, it's natural to use the original instrument and its square (unless the only instrument is a dummy, in which case you'll need a better idea).

You might be tempted, however, to work with a single first stage, say equation (4.6.2), and estimate the following second stage manually:

$$y_i = \alpha'X_i + \rho_1 \hat{s}_i + \rho_2 \hat{s}_i^2 + [\eta_i + \rho_1(s_i - \hat{s}_i) + \rho_2(s_i^2 - \hat{s}_i^2)].$$

This is a mistake, since $\hat{s}_i$ can be correlated with $s_i^2 - \hat{s}_i^2$ while $\hat{s}_i^2$ can be correlated with both $s_i - \hat{s}_i$ and $s_i^2 - \hat{s}_i^2$. In contrast, as long as X_i and Z_i are uncorrelated with η_i in (4.6.5) and you have enough instruments in Z_i , 2SLS estimation of (4.6.5) is straightforward.

4.6.2 Peer Effects

A vast literature in social science is concerned with peer effects. Loosely speaking, this means the causal effect of group characteristics on individual outcomes. Sometimes regression is used in an attempt to uncover these effects. In practice, the use of regression models to estimate peer effects is fraught with peril. Although this is not really an IV issue per se, the language and algebra of 2SLS help us understand why peer effects are hard to identify.

Broadly speaking, there are two types of peer effects. The first concerns the effect of group characteristics such as the average schooling in a state or city on individual outcomes as described by another variable. For example, Acemoglu and Angrist (2000) ask whether a given individual's earnings are affected by the average schooling in his or her state of residence. The theory of human capital externalities suggests that living in a state with a more educated workforce may make everyone in the state more productive, not just those who are more educated. This kind of spillover is said to be a *social return* to schooling: human capital that benefits everyone, whether or not they are more educated.

A causal model that allows for such externalities can be written

$$Y_{ijt} = \mu_j + \lambda_t + \gamma \bar{S}_{jt} + \rho s_i + u_{jt} + \eta_{ijt}, \quad (4.6.6)$$

where Y_{ijt} is the log weekly wage of individual i in state j in year t , u_{jt} is a state-year error component, and η_{ijt} is an individual error term. The controls μ_j and λ_t are state-of-residence and year effects. The coefficient ρ is the returns to schooling for an individual, while the coefficient γ is meant to capture the effect of average schooling, $\bar{S}_{jt}$, in state j and year t .

In addition to the usual concerns about s_i , the most important identification problem raised by equation (4.6.6) is omitted variables bias from correlation between average schooling and other state-year effects embodied in the error component u_{jt} . For example, public university systems may expand during cyclical upturns, generating a common trend in state average schooling levels and state average earnings. Acemoglu and Angrist (2000) attempt to solve this problem using instrumental variables derived from historical compulsory attendance laws that are correlated with $\bar{S}_{jt}$ but uncorrelated with contemporaneous u_{jt} and η_i .

While omitted state-year effects are the primary concern motivating Acemoglu and Angrist's (2000) IV estimation, the fact that one regressor, $\bar{S}_{jt}$, is the average of another regressor, s_i , also complicates the interpretation of OLS estimates of equation (4.6.6). To see this, consider a simpler version

of (4.6.6) with a cross-section dimension only. This can be written

$$Y_{ij} = \mu + \pi_0 s_i + \pi_1 \bar{S}_j + v_{ij}; \quad (4.6.7)$$

where Y_{ij} is the log weekly wage of individual i in state j and $\bar{S}_j$ is average schooling in the state. The coefficients π_0 and π_1 are defined so that the error, v_{ij} , is uncorrelated with both regressors. Now, let ρ_0 denote the coefficient from a bivariate regression of Y_{ij} on s_i only and let ρ_1 denote the coefficient from a bivariate regression of Y_{ij} on $\bar{S}_j$ only. From the discussion of grouping and 2SLS earlier in this chapter, we know that ρ_1 is the 2SLS estimate of the coefficient on s_i in a bivariate regression of Y_{ij} on s_i using a full set of state dummies as instruments. The appendix uses this fact to show that the parameters in equation (4.6.7) can be written in terms of ρ_0 and ρ_1 as

$$\begin{aligned} \pi_0 &= \rho_1 + \phi(\rho_0 - \rho_1) \\ \pi_1 &= \phi(\rho_1 - \rho_0), \end{aligned} \quad (4.6.8)$$

where $\phi = \frac{1}{1-R^2} > 1$, and R^2 is the first-stage R^2 when state dummies are used as instruments for s_i .

The upshot of (4.6.8) is that if, for any reason, OLS estimates of the bivariate regression of wages on individual schooling differ from 2SLS estimates using state dummy instruments, the coefficient on average schooling in (4.6.7) will be nonzero. For example, if instrumenting with state dummies corrects for attenuation bias due to measurement error in s_i , we have $\rho_1 > \rho_0$ and the spurious appearance of positive social returns. In contrast, if instrumenting with state dummies eliminates the bias from positive correlation between s_i and unobserved earnings potential, we have $\rho_1 < \rho_0$, and the appearance of negative social returns.³³ In practice, therefore, it is very difficult to isolate social effects by OLS estimation of

³³The coefficient on average schooling in an equation with individual schooling can be interpreted as the Hausman (1978) test statistic for the equality of OLS estimates and 2SLS estimates of private returns to schooling using state dummies as instruments. Borjas (1992) discusses a similar problem affecting the estimation of ethnic-background effects.

an equation like (4.6.6), though more sophisticated IV strategies where both the individual and group averages are treated as endogenous may work.

A second and even more difficult peer effect to uncover is the effect of the group average of a variable on the individual level of this same variable. This is not really an IV problem; it takes us back to basic regression issues. To see this point, suppose that $\bar{S}_j$ is the high school graduation rate in school j , and we would like to know whether students are more likely to graduate from high school when everyone around them is more likely to graduate from high school. To uncover the peer effect on high school graduation rates, we might work with a regression model like:

$$s_{ij} = \mu + \pi_2 \bar{S}_j + v_{ij}, \quad (4.6.9)$$

where s_{ij} is individual i 's high school graduation status and $\bar{S}_j$ is the average high school graduation rate in school j , which i attends.

At first blush, equation (4.6.9) seems like a sensible formulation of a well-defined causal question, but in fact it is nonsense. The regression of s_{ij} on $\bar{S}_j$ *always* has a coefficient of 1, a conclusion that can be drawn immediately once you recognize $\bar{S}_j$ as the first-stage fitted value from a regression of s_{ij} on a full set of school dummies.³⁴ Thus, an equation like (4.6.9) cannot possibly be informative about causal effects. A modestly improved version of this bad peer regression changes (4.6.9) to

$$s_{ij} = \mu + \pi_3 \bar{S}_{(i)j} + v_{ij}, \quad (4.6.10)$$

³⁴Here is a direct proof that the regression of s_{ij} on $\bar{S}_j$ is always unity:

$$\begin{aligned} \frac{\sum_j \sum_i s_{ij}(\bar{S}_j - \bar{S})}{\sum_j n_j(\bar{S}_j - \bar{S})^2} &= \frac{\sum_j (\bar{S}_j - \bar{S}) \sum_i s_{ij}}{\sum_j n_j(\bar{S}_j - \bar{S})^2} \\ &= \frac{\sum_j (\bar{S}_j - \bar{S})(n_j \bar{S}_j)}{\sum_j n_j(\bar{S}_j - \bar{S})^2} = 1. \end{aligned}$$

where $\bar{s}_{(i)j}$ is the mean of s_{ij} in school j , excluding student i . This is a step in the right direction— π_3 is no longer automatically equal to 1—but still problematic because s_{ij} and $\bar{s}_{(i)j}$ are both affected by school-level random shocks that are implicitly part of v_{ij} . The presence of random group effects in the error term raises important issues for statistical inference, issues discussed at length in chapter 8. But in an equation like (4.6.10), group-level random shocks are more than a problem for standard errors: any shock common to the group (school) creates spurious peer effects. For example, particularly effective school principals may raise graduation rates for everyone in the schools at which they work. This looks like a peer effect, since it induces correlation between s_{ij} and $\bar{s}_{(i)j}$ even if there is no causal link between peer means and individual student achievement. We therefore prefer not to see regressions like (4.6.10) either.

The best shot at a causal investigation of peer effects focuses on variation in ex ante peer characteristics, that is, some measure of peer quality that predates the outcome variable and is therefore unaffected by common shocks. A recent example is Ammermueller and Pischke (2006), who studied the link between classmates' family background, as measured by the number of books in their homes, and student achievement in European primary schools. The Ammermueller and Pischke regressions are versions of

$$s_{ij} = \mu + \pi_4 \bar{B}_{(i)j} + v_{ij},$$

where $\bar{B}_{(i)j}$ is the average number of books in the home of student i 's peers. This looks like (4.6.10), but with an important difference. The variable $\bar{B}_{(i)j}$ is a feature of the home environment that predates test scores and is therefore unaffected by school-level random shocks.

Angrist and Lang (2004) provide another example of an attempt to link student achievement with the ex ante characteristics of peers. The Angrist and Lang study looked at the impact of bused-in low-achieving newcomers on high-achieving residents' test scores. The regression of interest in

this case is a version of

$$s_{ij} = \mu + \pi_5 \bar{m}_j + v_{ij}, \quad (4.6.11)$$

where $\bar{m}_j$ is the number of bused-in low achievers in school j and s_{ij} is resident student i 's test score. Spurious correlation due to common shocks is not a concern in this context, for two reasons. First, $\bar{m}_j$ is a feature of the school population determined by students outside the sample used to estimate (4.6.11). Second, the number of low achievers is an *ex ante* variable biased on information about where the students come from and not the outcome variable, s_{ij} . School-level random effects that are part of v_{ij} remain an important issue for inference, however, since $\bar{m}_j$ is a group-level variable.

4.6.3 *Limited Dependent Variables Reprise*

In section 3.4.2, we discussed the consequences of limited dependent variables for regression models. When the dependent variable is binary or non-negative—say, employment status or hours worked—the CEF is typically nonlinear. Most nonlinear LDV models are built around a nonlinear transformation of a linear latent index. Examples include probit, logit, and Tobit. These models capture features of the associated CEFs (e.g., probit fitted values are guaranteed to be between zero and one, while Tobit fitted values are non-negative). Yet we saw that the added complexity and extra work required to interpret the results from latent index models may not be worth the trouble.

An important consideration in favor of OLS is a conceptual robustness that structural models often lack. OLS always gives a MMSE linear approximation to the CEF. In fact, we can think of OLS as a scheme for computing marginal effects—a scheme that has the virtue of simplicity, automation, and comparability across studies. Nonlinear latent index models are more like GLS: they provide an efficiency gain when taken literally, but require a commitment to functional form and distributional assumptions, about which we do not usually feel

strongly.³⁵ A second consideration is the distinction between the latent index parameters at the heart of nonlinear models and the average causal effects that we believe should be the objects of primary interest in most research projects.

The arguments in favor of conventional OLS with LDVs apply with equal force to 2SLS and models with endogenous variables. IV methods capture local average treatment effects regardless of whether the dependent variable is binary, non-negative, or continuously distributed. With covariates, we can think of 2SLS as estimating LATE averaged across covariate cells. In models with variable or continuous treatment intensity, 2SLS gives us the average causal response or an average derivative. Although Abadie (2003) has shown that 2SLS does not, in general, provide the MMSE approximation to the complier causal response function, in practice, 2SLS estimates come out remarkably close to estimates using the more rigorously grounded Abadie procedure (and with a saturated model for covariates, 2SLS and Abadie are the same). Moreover, 2SLS estimates LATE directly; there is no intermediate step involving the calculation of marginal effects.

2SLS is not the only way to go. An alternative, more elaborate approach tries to build up a causal story by describing the process generating LDVs in detail. A good example is bivariate probit, which can be applied to the Angrist and Evans (1998) example like this. Suppose that a woman decides to

³⁵The analogy between nonlinear LDV models and GLS is more than rhetorical. Consider a probit model with nonlinear CEF, $E[y_i|X_i] = \Phi\left[\frac{X_i'\beta^*}{\sigma}\right] \equiv r_i$. The first-order conditions for maximum likelihood estimation of this model are

$$\sum_i \frac{(y_i - r_i)X_i}{r_i(1 - r_i)} = 0.$$

Maximum likelihood is asymptotically the same as GLS estimation of the nonlinear regression model

$$y_i = \Phi\left[\frac{X_i'\beta^*}{\sigma}\right] + \xi_j,$$

since the conditional variance of y_i is $r_i(1 - r_i)$. The only difference is that GLS is done in two steps.

have a third child by comparing costs and benefits using a net benefit function or latent index that is linear in covariates and excluded instruments, with a random error term, v_i . The bivariate probit first stage can be written

$$D_i = 1[X_i' \gamma_0^* + \gamma_1^* Z_i > v_i], \quad (4.6.12)$$

where Z_i is an instrumental variable that increases the benefit of a third child, conditional on covariates, X_i . For example, American parents appear to value a third child more when they have had either two boys or two girls, a sort of portfolio diversification phenomenon that can be understood as increasing the benefit of a third child in families with same-sex sibships.

An outcome variable of primary interest in this context is employment status, a Bernoulli random variable with a conditional mean between zero and one. To complete the model, suppose that employment status, y_i , is determined by the latent index

$$y_i = 1[X_i' \beta_0^* + \beta_1^* D_i > \varepsilon_i], \quad (4.6.13)$$

where ε_i is a second random component or error term. This latent index can be seen as arising from a comparison of the costs and benefits of working.

The source of omitted variables bias in the bivariate probit setup is correlation between v_i and ε_i . In other words, unmeasured random determinants of childbearing are correlated with unmeasured random determinants of employment. The model is identified by assuming Z_i is independent of these components, and that the random components are normally distributed. Given normality, the parameters in (4.6.12) and (4.6.13) can be estimated by maximum likelihood. The log likelihood function is

$$\begin{aligned} \sum y_i \ln \Phi_b \left(\frac{X_i' \beta_0^* + \beta_1^* D_i}{\sigma_\varepsilon}, \frac{X_i' \gamma_0^* + \gamma_1^* Z_i}{\sigma_v}; \rho_{\varepsilon v} \right) \\ + (1 - y_i) \ln \left[1 - \Phi_b \left(\frac{X_i' \beta_0^* + \beta_1^* D_i}{\sigma_\varepsilon}, \frac{X_i' \gamma_0^* + \gamma_1^* Z_i}{\sigma_v}; \rho_{\varepsilon v} \right) \right], \end{aligned} \quad (4.6.14)$$

where $\Phi_b(\cdot, \cdot; \rho_{\varepsilon v})$ is the bivariate normal distribution function with correlation coefficient $\rho_{\varepsilon v}$. Note, however, that we can multiply the latent index coefficients and error standard deviations $(\sigma_\varepsilon, \sigma_v)$ by a positive constant without changing the likelihood. The object of estimation is therefore the ratio of the index coefficients to the error standard deviations (e.g., $\beta_1^*/\sigma_\varepsilon$).

The potential outcomes defined by the bivariate probit model are

$$y_{0i} = 1[X_i'\beta_0^* > \varepsilon_i] \text{ and } y_{1i} = 1[X_i'\beta_0^* + \beta_1^* > \varepsilon_i],$$

while potential treatment assignments are

$$D_{0i} = 1[X_i'\gamma_0^* > v_i] \text{ and } D_{1i} = 1[X_i'\gamma_0^* + \gamma_1^* > v_i].$$

As usual, only one potential outcome and one potential assignment are observed for any one person. It's also clear from this representation that correlation between v_i and ε_i is the same thing as correlation between potential treatment assignments and potential outcomes.

The latent index coefficients do not themselves tell us anything about the size of the causal effect of childbearing on employment other than the sign. To see this, note that the average causal effect of childbearing is

$$E[y_{1i} - y_{0i}] = E\{1[X_i'\beta_0^* + \beta_1^* > \varepsilon_i] - 1[X_i'\beta_0^* > \varepsilon_i]\},$$

while the average effect on the treated is

$$\begin{aligned} E[y_{1i} - y_{0i} | D_i = 1] \\ = E\{1[X_i'\beta_0^* + \beta_1^* > \varepsilon_i] - 1[X_i'\beta_0^* > \varepsilon_i] | X_i'\gamma_0^* + \gamma_1^* Z_i > v_i\}. \end{aligned}$$

Given alternative distributional assumptions for v_i and ε_i , these can be anything. (If the error terms are heteroskedastic, then even the sign of these expressions is indeterminate.)

By virtue of normality, the average causal effects generated by the bivariate probit model are easy to evaluate. The average treatment effect is

$$\begin{aligned} & E\{1[X'_i\beta_0^* + \beta_1^* > \varepsilon_i] - 1[X'_i\beta_0^* > \varepsilon_i]\} \\ &= E\left\{\Phi\left[\frac{X'_i\beta_0^* + \beta_1^*}{\sigma_\varepsilon}\right] - \Phi\left[\frac{X'_i\beta_0^*}{\sigma_\varepsilon}\right]\right\}, \end{aligned} \quad (4.6.15)$$

where $\Phi[\cdot]$ is the normal CDF. The effect on the treated is a little more complicated since it involves the bivariate normal CDF:

$$\begin{aligned} & E[Y_{1i} - Y_{0i} | D_i = 1] \\ &= E\left\{\frac{\Phi_b\left(\frac{X'_i\beta_0^* + \beta_1^*}{\sigma_\varepsilon}, \frac{X'_i\gamma_0^* + \gamma_1^*Z_i}{\sigma_v}; \rho_{\varepsilon v}\right) - \Phi_b\left(\frac{X'_i\beta_0^*}{\sigma_\varepsilon}, \frac{X'_i\gamma_0^* + \gamma_1^*Z_i}{\sigma_v}; \rho_{\varepsilon v}\right)}{\Phi\left(\frac{X'_i\gamma_0^* + \gamma_1^*Z_i}{\sigma_v}\right)}\right\}. \end{aligned} \quad (4.6.16)$$

The bivariate normal CDF is a canned function in many software packages, so this is easy enough to calculate in practice.

Bivariate probit probably qualifies as harmless in the sense that it's not very complicated and easy to get right using packaged software routines. Still, it shares the disadvantages of nonlinear latent index modeling discussed in section 3.4.2. First, some researchers become distracted by an effort to estimate index coefficients instead of average causal effects. For example, a large literature in econometrics is concerned with the estimation of index coefficients without the need for distributional assumptions. Applied researchers interested in causal effects can safely ignore this work.³⁶

³⁶Suppose the latent error term has an unknown distribution, with CDF $\Lambda[\cdot]$. The average causal effect in this case is

$$E\{\Lambda[X'_i\beta_0^* + \beta_1^*] - \Lambda[X'_i\beta_0^*]\} = \Lambda'[X'_i\beta_0^* + \tilde{\beta}_1]\beta_1^*,$$

where (by the mean value theorem) $\tilde{\beta}_1$ is a number in $[0, \beta_1^*]$. This always depends on the shape of $\Lambda[\cdot]$, so it is never enough to know the index coefficients alone.

A second vice in this context is also a virtue. Bivariate probit and other models of this sort can be used to estimate unconditional average causal effects and/or effects on the treated. In contrast, 2SLS does not promise you average causal effects, only *local* average causal effects. But it should be clear from (4.6.15) that the assumed normality of the latent index error terms is essential for this. As always, the best you can do without a distributional assumption is LATE, the average causal effect for compliers. For bivariate probit, we can write LATE as

$$\begin{aligned} E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] \\ &= E\{1[X'_i\beta_0^* + \beta_1^* > \varepsilon_i] \\ &\quad - 1[X'_i\beta_0^* > \varepsilon_i] | X'_i\gamma_0^* + \gamma_1^* > v_i > X'_i\gamma_0^*\}, \end{aligned}$$

which, like (4.6.16), can be evaluated using joint normality of v_i and ε_i . But you needn't bother using normality to evaluate $E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]$, since LATE can be estimated by IV for each X_i and averaged using the histogram of the covariates. Alternately, do 2SLS and settle for a variance-weighted average of covariate-specific LATEs, as described by the saturate and weight theorem in section 4.5.3.

You might be wondering whether LATE is enough. Perhaps you would like to estimate the unconditional average treatment effect or the effect of treatment on the treated and are willing to make a few extra assumptions to do so. That's all well and good, but in our experience you can't get blood from a stone, even with heroic assumptions. Since local information is all that's in the data, in practice, the average causal effects produced by bivariate probit are likely to be similar to 2SLS estimates, provided the model for covariates is sufficiently flexible. This is illustrated in table 4.6.1, which reports 2SLS and bivariate probit estimates of the effects of a third child on female labor supply using the Angrist-Evans (1998) same-sex instruments and the same 1980 census sample of married women with two or more children used in their paper. The dependent variable is a dummy for having worked the previous year; the endogenous variable is a dummy for

TABLE 4.6.1
2SLS, Abadie, and bivariate probit estimates of the effects of a third
child on female labor supply

2SLS (1)	Abadie Estimates		Bivariate Probit		
	Linear (2)	Probit (3)	MFX (4)	ATE (5)	TOT (6)
A. No Covariates					
-.138 (.029)	-.138 (.030)	-.137 (.030)	-.138 (.029)	-.139 (.029)	-.139 (.029)
B. Some covariates (no age controls)					
-.132 (.029)	-.132 (.029)	-.131 (.028)	-.135 (.028)	-.135 (.028)	-.135 (.028)
C. Some covariates plus age at first birth					
-.129 (.028)	-.129 (.028)	-.129 (.028)	-.133 (.026)	-.133 (.026)	-.133 (.026)
D. Some covariates plus age at first birth and a dummy for age > 30					
-.124 (.028)	-.125 (.029)	-.125 (.029)	-.131 (.025)	-.131 (.025)	-.131 (.025)
E. Some covariates plus age at first birth and age					
-.120 (.028)	-.121 (.026)	-.121 (.026)	-.171 (.023)	-.171 (.023)	-.171 (.023)

Notes: Adapted from Angrist (2001). The table compares 2SLS estimates to alternative estimates of the effect of childbearing on labor supply using nonlinear models. All models use same-sex instruments. Standard errors for the Abadie estimates were bootstrapped using 100 replications of subsamples of size 20,000. MFX denotes marginal effects; ATE is the unconditional average treatment effect; TOT is the average effect of treatment on the treated.

having a third child. The first-stage effect of a same-sex sibship on the probability of a third birth is about 7 percentage points.

Panel A of table 4.6.1 reports estimates from a model with no covariates. The 2SLS estimate of $-.138$ in column 1 is numerically identical to the Abadie causal effect estimated using a linear model in column 2, as it should be in this case. Without covariates, the 2SLS slope coefficient provides the best linear approximation to the complier causal response

function as does Abadie's kappa-weighting procedure. The marginal effect changes little if, instead of a linear approximation, we use nonlinear least squares with a probit CEF. The marginal effect estimated by minimizing

$$E \left\{ \kappa_i \left(y_i - \Phi \left[\frac{\beta_0^* + \beta_1^* D_i}{\sigma_\varepsilon} \right] \right)^2 \right\}$$

is $-.137$, reported in column 3. This is not surprising, since the model without covariates imposes no functional form assumptions.

Perhaps more surprising is the fact that marginal effects and the average treatment effects calculated using (4.6.15) and (4.6.16) are also the same as the 2SLS and Abadie estimates. These results are reported in columns 4–6. The marginal effect calculated using a derivative to approximate to the finite difference in (4.6.15) is $-.138$ (in column 4, labeled MFX for marginal effects), while both average treatment effects are $-.139$ in columns 5 and 6. Adding a few covariates has little effect on the estimates, as can be seen in panel B. In this case, the covariates are all dummy variables, three for race (black, Hispanic, other), and two indicating first- and second-born boys (the excluded instrument is the interaction of these two). Panels C and D show that adding a linear term in age at first birth and a dummy for maternal age also leaves the estimates unchanged.

The invariance to covariates seems desirable: because the same-sex instrument is essentially independent of covariates, control for covariates is unnecessary to eliminate bias and should primarily affect precision. Yet, as panel E shows, the marginal effects generated by bivariate probit are sensitive to the list of covariates. Swapping a dummy indicating mothers over 30 with a linear age term increases the bivariate probit estimates markedly, to $-.171$, while leaving 2SLS and the Abadie estimators unchanged. This probably reflects the fact that the linear age term induces an extrapolation into cells where there is little data. Although there is no harm in reporting the bivariate probit effects in panel E, it's hard to

see why the more robust 2SLS and Abadie estimators should not be preferred.³⁷

4.6.4 *The Bias of 2SLS*★

It is a fortunate fact that the OLS estimator is not only consistent, it is also unbiased (as we briefly noted at the end of section 3.1.3). This means that in a sample of any size, the estimated OLS coefficient vector has a distribution that is centered on the population coefficient vector.³⁸ The 2SLS estimator, in contrast, is consistent, but biased. This means that the 2SLS estimator only promises to be close to the causal effect of interest in large samples. In small samples, 2SLS estimates can differ systematically from the target parameter.

For many years, applied researchers lived with the knowledge that 2SLS is biased without losing too much sleep. Neither of us heard much about the bias of 2SLS in our graduate econometrics classes. A series of papers in the early 1990s changed this, however. These papers show that 2SLS estimates can be highly misleading in cases relevant for empirical practice.³⁹

The 2SLS estimator is most biased when the instruments are “weak,” meaning the correlation with endogenous regressors is low, and when there are many overidentifying restrictions. When the instruments are both many and weak, the 2SLS estimator is biased toward the probability limit of the corresponding OLS estimate. In the worst-case scenario, when the instruments are so weak that there is no first stage in the population, the 2SLS sampling distribution is centered on the

³⁷Angrist (2001) makes the same point using twins instruments and reports a similar pattern in a comparison of 2SLS, Abadie, and nonlinear structural estimates of models for hours worked.

³⁸A more precise statement is that OLS is unbiased when either (1) the CEF is linear or (2) the regressors are nonstochastic, that is, fixed in repeated samples. In practice, these qualifications do not seem to matter much. As a rule, the sampling distribution of $\hat{\beta} = [\sum_i X_i X_i']^{-1} \sum_i X_i Y_i$, tends to be centered on the population analog, $\beta = E[X_i X_i']^{-1} E[X_i Y_i]$, in samples of any size, whether or not the CEF is linear or the regressors are stochastic.

³⁹Key references are Nelson and Startz (1990a,b), Buse (1992), Bekker (1994), and especially Bound, Jaeger, and Baker (1995).

probability limit of OLS. The theory behind this result is a little technical, but the basic idea is easy to see. The source of the bias in 2SLS estimates is the randomness in estimates of the first-stage fitted values. In practice, the first-stage estimates reflect some of the randomness in the endogenous variable, since the first-stage coefficients come from a regression of the endogenous variable on the instruments. If the population first stage is zero, then all randomness in the first stage is due to the endogenous variable. This randomness generates finite-sample correlation between first-stage fitted values and second-stage errors, since the endogenous variable is correlated with the second-stage errors (or else you wouldn't be instrumenting in the first place).

A more formal derivation of 2SLS bias goes like this. To streamline the discussion we use matrices and vectors and a simple constant-effects model (it's difficult to discuss bias in a heterogeneous effects world, since the target parameter may change as the number of instruments changes). Suppose you are interested in estimating the effect of a single endogenous regressor, stored in a vector x , on a dependent variable, stored in the vector y , with no other covariates. The causal model of interest can then be written

$$y = \beta x + \eta. \quad (4.6.17)$$

The $N \times Q$ matrix of instrumental variables is Z , with the associated first-stage equation

$$x = Z\pi + \xi. \quad (4.6.18)$$

OLS estimates of (4.6.17) are biased because η_i is correlated with ξ_i . The instruments Z_i are uncorrelated with ξ_i by construction and uncorrelated with η_i by assumption.

The 2SLS estimator is

$$\hat{\beta}_{2SLS} = (x'P_Zx)^{-1}x'P_Zy = \beta + (x'P_Zx)^{-1}x'P_Z\eta,$$

where $P_Z = Z(Z'Z)^{-1}Z'$ is the projection matrix that produces fitted values from a regression of x on Z . Substituting for x in

$x'P_Z\eta$, we get

$$\begin{aligned}\hat{\beta}_{2SLS} - \beta &= (x'P_Zx)^{-1}(\pi'Z' + \xi')P_Z\eta \\ &= (x'P_Zx)^{-1}\pi'Z'\eta + (x'P_Zx)^{-1}\xi'P_Z\eta.\end{aligned}\quad (4.6.19)$$

The bias in 2SLS comes from the nonzero expectation of terms on the right-hand side.

The expectation of (4.6.19) is hard to evaluate because the expectation operator does not pass through the inverse $(x'P_Zx)^{-1}$, a nonlinear function. It's possible to show, however, that the expectation of the ratios on the right-hand side of (4.6.19) can be closely approximated by the ratio of expectations. In other words,

$$E[\hat{\beta}_{2SLS} - \beta] \approx (E[x'P_Zx])^{-1}E[\pi'Z'\eta] + (E[x'P_Zx])^{-1}E[\xi'P_Z\eta].$$

This approximation is much better than the usual asymptotic approximation invoked in large-sample theory, so we think of it as giving us a good measure of the finite-sample behavior of the 2SLS estimator.⁴⁰ Furthermore, because $E[\pi'Z'\xi] = 0$ and $E[\pi'Z'\eta] = 0$, we have

$$E[\hat{\beta}_{2SLS} - \beta] \approx [E(\pi'Z'Z\pi) + E(\xi'P_Z\xi)]^{-1}E(\xi'P_Z\eta).\quad (4.6.20)$$

The approximate bias of 2SLS therefore comes from the fact that $E(\xi'P_Z\eta)$ is not zero unless η_i and ξ_i are uncorrelated. But correlation between η_i and ξ_i is what led us to use IV in the first place.

Further manipulation of (4.6.20) generates an expression that is especially useful:

$$E[\hat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\eta\xi}}{\sigma_\xi^2} \left[\frac{E(\pi'Z'Z\pi)/Q}{\sigma_\xi^2} + 1 \right]^{-1}$$

⁴⁰See Bekker (1994) and Angrist and Krueger (1995). This is also called a group-asymptotic approximation because it can be derived from an asymptotic sequence that lets the number of instruments go to infinity at the same time as the number of observations goes to infinity, keeping the number of observations per instrument (group) constant.

(see the appendix for a derivation). The term $(1/\sigma_\xi^2)E(\pi'Z'Z\pi)/Q$ is the F -statistic for the joint significance of all regressors in the first stage regression.⁴¹ Call this statistic F , so that we can write

$$E[\hat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\eta\xi}}{\sigma_\xi^2} \frac{1}{F + 1}. \quad (4.6.21)$$

From this we see that as the first stage F -statistic gets small, the bias of 2SLS approaches $\sigma_{\eta\xi}/\sigma_\xi^2$. The bias of the OLS estimator is $\sigma_{\eta\xi}/\sigma_x^2$, which also equals $\sigma_{\eta\xi}/\sigma_\xi^2$ if $\pi = 0$. Thus, we have shown that 2SLS is centered on the same point as OLS when the first stage is zero. More generally, we can say 2SLS estimates are “biased toward OLS estimates” when there isn’t much of a first stage. On the other hand, the bias of 2SLS vanishes when F gets large, as should happen in large samples when $\pi \neq 0$.

When the instruments are weak, the F -statistic varies inversely with the number of instruments. To see why, consider adding useless instruments to your 2SLS model, that is, instruments with no effect on the first-stage R^2 . The model sum of squares, $E(\pi'Z'Z\pi)$, and the residual variance, σ_ξ^2 , will both stay the same while Q goes up. The F -statistic becomes smaller as a result. From this we learn that the addition of many weak instruments increases bias.

Intuitively, the bias in 2SLS is a consequence of the fact that the first-stage is estimated. If the first stage coefficients were known, we could use $\hat{x}_{pop} = Z\pi$ for the first-stage fitted values. These fitted values are uncorrelated with the second-stage error. In practice, however, we use $\hat{x} = P_Z x = Z\pi + P_Z \xi$, which differs from $\hat{x}_{pop}$ by the term $P_Z \xi$. The bias in 2SLS arises from the fact that $P_Z \xi$ is correlated with η , so some of

⁴¹Sort of; the actual F -statistic is $(1/\hat{\sigma}_\xi^2)\hat{\pi}'Z'Z\hat{\pi}/Q$, where hats denote estimates. $(1/\sigma_\xi^2)E(\pi'Z'Z\pi)/Q$ is therefore sometimes called the population F -statistic since it’s the F -statistic we’d get in an infinitely large sample. In practice, the distinction between population and sample F matters little in this context. Some econometricians prefer to multiply the first-stage F by the number of instruments when summarizing instrument strength. This product is called the “concentration parameter.”

the correlation between errors in the first and second stages seeps into our 2SLS estimates through the sampling variability in $\hat{\pi}$. Asymptotically, this correlation disappears, but real life does not play out in asymptopia.

Formula (4.6.21) shows that, other things equal, the bias in 2SLS is an increasing function of the number of instruments, so bias is least in the just-identified case when the number of instruments is as low as it can get. In fact, just-identified 2SLS (say, the simple Wald estimator) is approximately *unbiased*. This is hard to show formally because just-identified 2SLS has no moments (i.e., the sampling distribution has fat tails). Nevertheless, even with weak instruments, just-identified 2SLS is approximately centered where it should be. We therefore say that just-identified 2SLS is median-unbiased. This is not to say that you can happily use weak instruments in just-identified models. With a weak instrument, just-identified estimates tend to be too imprecise to be useful.

The limited information maximum likelihood (LIML) estimator is approximately median-unbiased for overidentified constant effects models, and therefore provides an attractive alternative to just-identified estimation using one instrument at a time (see, e.g., Davidson and MacKinnon, 1993, and Mariano, 2001). LIML has the advantage of having the same asymptotic distribution as 2SLS (under constant effects) while providing a finite-sample bias reduction. A number of other estimators also reduce the bias in overidentified 2SLS models. But an extensive Monte Carlo study by Flores-Lagunes (2007) suggests that LIML does at least as well as the alternatives in a wide range of circumstances (in terms of bias, mean absolute error, and the empirical rejection rates for t -tests). Another advantage of LIML is that many statistical packages compute it, while other estimators typically require some programming.⁴²

⁴²LIML is available in SAS and in STATA 10. With weak instruments, LIML standard errors are not quite right, but Bekker (1994) gives a simple fix for this. Why is LIML unbiased? Expression (4.6.21) shows that the approximate bias of 2SLS is proportional to the bias of OLS. From this we conclude that there is a linear combination of OLS and 2SLS that is approximately

We use a small Monte Carlo experiment to illustrate some of the theoretical results from the discussion above. The simulated data are drawn from the following model,

$$y_i = \beta x_i + \eta_i$$

$$x_i = \sum_{j=1}^Q \pi_j z_{ij} + \xi_i,$$

with $\beta = 1$, $\pi_1 = 0.1$, $\pi_j = 0$ for $j = 2, \dots, Q$; and

$$\begin{pmatrix} \eta_i \\ \xi_i \end{pmatrix} \Big| Z \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} \right),$$

where the z_{ij} are independent, normally distributed random variables with mean zero and unit variance. This simulates a scenario with one good instrument and $Q - 1$ worthless instruments. The sample size is 1000.

Figure 4.6.1 shows the Monte Carlo cumulative distribution functions of four estimators: OLS, just-identified IV (i.e., 2SLS with $Q = 1$, labeled IV; first-stage $F = 11.1$), 2SLS with two instruments ($Q = 2$, labeled 2SLS; first-stage $F = 6.0$), and LIML with $Q = 2$. The OLS estimator is biased and centered around a value of about 1.79. IV is centered around 1, the value of β . 2SLS with one weak and one uninformative instrument is moderately biased toward OLS (the median is 1.07). The distribution function for LIML with $Q = 2$ is indistinguishable from that for just-identified IV, even though the LIML estimator also uses an uninformative instrument.

Figure 4.6.2 reports simulation results where we set $Q = 20$. Thus, in addition to the one informative but weak instrument,

unbiased. LIML turns out to be just such a “combination estimator.” Like the bias of 2SLS, the approximate unbiasedness of LIML can be shown using a Bekker-style group-asymptotic sequence that fixes the ratio of instruments to sample size. It is worth mentioning, however, that LIML is biased in models with a certain type of heteroskedasticity. See Bekker and van der Ploeg (2005) and Hausman, et al. (2008) for details. Unlike LIML, the Jackknife IV Estimator (JIVE: see, e.g., Angrist, Imbens, and Krueger, 1999) is Bekker-unbiased under heteroskedasticity. Akerberg and Devereux (2007) recently introduced an improved version of JIVE with lower variance.

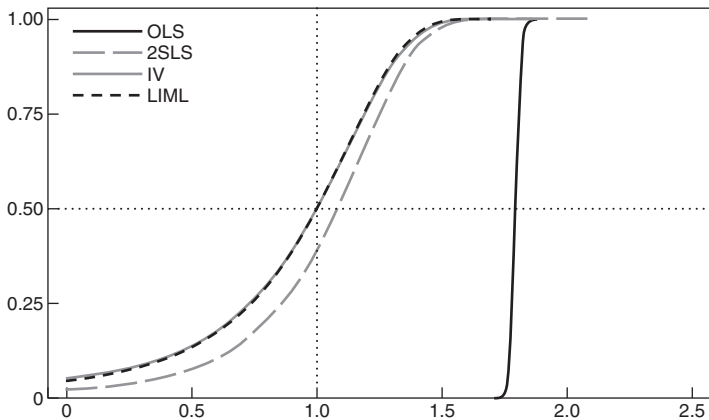


Figure 4.6.1 Monte Carlo cumulative distribution functions of OLS, IV ($Q = 1$), 2SLS ($Q = 2$), and LIML ($Q = 2$) estimators.

we added 19 worthless instruments (first-stage $F = 1.51$). The figure again shows OLS, 2SLS, and LIML distributions. The bias in 2SLS is now much worse (the median is 1.53, close to the OLS median). The sampling distribution of the 2SLS estimator is also much tighter than in the $Q = 2$ case. LIML

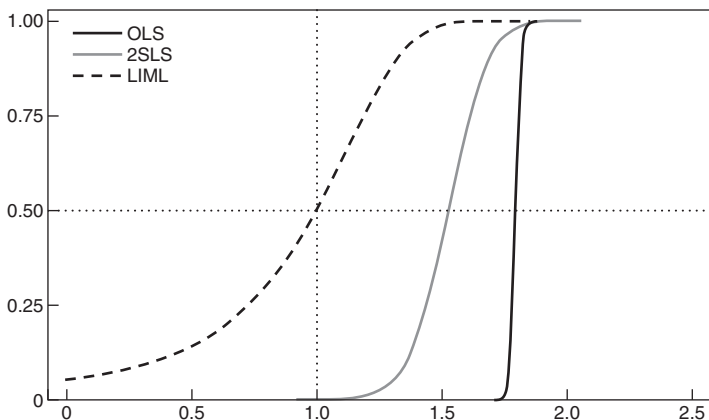


Figure 4.6.2 Monte Carlo cumulative distribution functions of OLS, 2SLS, and LIML estimators with $Q = 20$ instruments.

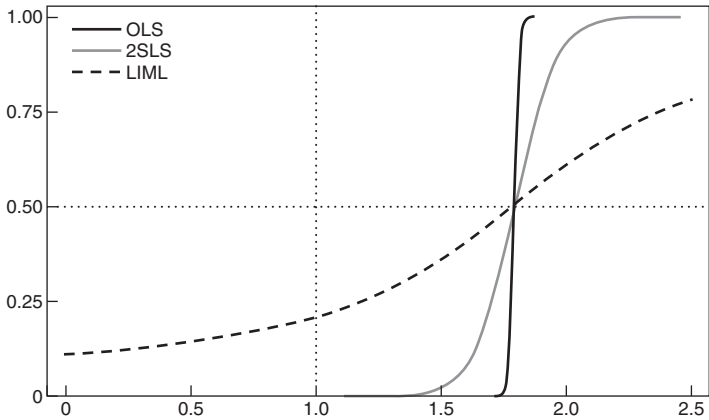


Figure 4.6.3 Monte Carlo cumulative distribution functions of OLS, 2SLS, and LIML estimators with $Q = 20$ worthless instruments.

again performs well and is centered around $\beta = 1$, with a bit more dispersion than in the $Q = 2$ case.

Finally, figure 4.6.3 reports simulation results from a model that is truly unidentified. In this case, we set $\pi_j = 0$; $j = 1, \dots, 20$ (first-stage $F = 1.0$). Not surprisingly, all the sampling distributions are centered around the same value as OLS. On the other hand, the 2SLS sampling distribution is much tighter than the LIML distribution. We would say advantage LIML in this case because the widely dispersed LIML sampling distribution correctly reflects the fact that the data are uninformative about the parameter of interest.

What does this mean in practice? Besides retaining a vague sense of worry about your first stage, we recommend the following:

1. Report the first stage and think about whether it makes sense. Are the magnitude and sign as you would expect, or are the estimates too big or wrong-signed? If so, perhaps your hypothesized first-stage mechanism isn't really there, rather, you simply got lucky.

2. Report the F -statistic on the excluded instruments. The bigger this is, the better. Stock, Wright, and Yogo (2002) suggest that F -statistics above about 10 put you in the safe zone, though obviously this cannot be a theorem.
3. Pick your single best instrument and report just-identified estimates using this one only. Just-identified IV is median-unbiased and therefore unlikely to be subject to a weak instruments critique.
4. Check overidentified 2SLS estimates with LIML. LIML is less precise than 2SLS but also less biased. If the results come out similar, be happy. If not, worry, and try to find stronger instruments or reduce the degree of overidentification.
5. Look at the coefficients, t -statistics, and F -statistics for excluded instruments in the reduced-form regression of dependent variables on instruments. Remember that the reduced form is proportional to the causal effect of interest. Moreover, the reduced-form estimates, since they are OLS, are unbiased. As Angrist and Krueger (2001) note, if you can't see the causal relation of interest in the reduced form, it's probably not there.⁴³

We illustrate some of this reasoning in a reanalysis of data from the Angrist and Krueger (1991) quarter-of-birth study. Bound, Jaeger, and Baker (1995) argued that bias is a major concern when using quarter of birth as an instrument for schooling, even though the sample size exceeds 300,000. ("Small sample" is clearly relative.) Earlier in the chapter, we saw that the quarter-of-birth pattern in schooling is reflected in the reduced form, so there would seem to be little cause for concern. On the other hand, Bound, Jaeger, and Baker (1995) argue that the most relevant models have additional controls not included in these reduced forms. Table 4.6.2 reproduces some of the specifications from Angrist and Krueger (1991) as well as other specifications in the spirit of Bound, Jaeger, and Baker (1995).

⁴³A recent paper by Chernozhukov and Hansen (2008) formalizes this maxim.

TABLE 4.6.2
Alternative IV estimates of the economic returns to schooling

	(1)	(2)	(3)	(4)	(5)	(6)
2SLS	.105 (.020)	.435 (.450)	.089 (.016)	.076 (.029)	.093 (.009)	.091 (.011)
LIML	.106 (.020)	.539 (.627)	.093 (.018)	.081 (.041)	.106 (.012)	.110 (.015)
<i>F</i> -statistic (excluded instruments)	32.27	.42	4.91	1.61	2.58	1.97
<i>Controls</i>						
Year of birth	✓	✓	✓	✓	✓	✓
State of birth					✓	✓
Age, age squared		✓		✓		✓
<i>Excluded instruments</i>						
Quarter-of-birth dummies	✓	✓				
Quarter of birth*year of birth			✓	✓	✓	✓
Quarter of birth*state of birth					✓	✓
Number of excluded instruments	3	2	30	28	180	178

Notes: The table compares 2SLS and LIML estimates using alternative sets of instruments and controls. The age and age squared variables measure age in quarters. The OLS estimate corresponding to the models reported in columns 1–4 is .071; the OLS estimate corresponding to the models reported in columns 5 and 6 is .067. Data are from the Angrist and Krueger (1991) 1980 census sample. The sample size is 329,509. Standard errors are reported in parentheses.

The first column in the table reports 2SLS and LIML estimates of a model using three quarter-of-birth dummies as instruments, with year-of-birth dummies as covariates. The OLS estimate for this specification is 0.071, while the 2SLS estimate is a bit higher, at 0.105. The first-stage *F*-statistic is over 32, well out of the danger zone. Not surprisingly, the LIML estimate is almost identical to 2SLS in this case.

Angrist and Krueger (1991) experimented with models that include age and age squared measured in quarters as additional controls. These controls are meant to pick up omitted age effects that might confound the quarter-of-birth instruments. The addition of age and age squared reduces the number of instruments to two, since age in quarters, year of birth, and quarter of birth are linearly dependent. As shown in column 2, the first-stage *F*-statistic drops to 0.4 when age and age squared are included as controls, a sure sign of trouble. But the 2SLS

standard error is high enough that we would not draw any substantive conclusions from this estimate. The LIML estimate is even less precise. This model is effectively unidentified.

Columns 3 and 4 report the results of adding interactions between quarter-of-birth dummies and year-of-birth dummies to the instrument list, so that there are 30 instruments, or 28 when the age and age squared variables are included. The first-stage F -statistics are 4.9 and 1.6 in these two specifications. The 2SLS estimates are a bit lower than in column 1 and hence closer to OLS. But LIML is not too far away from 2SLS. Although the LIML standard error is pretty big in column 4, it is not so large that the estimate is uninformative. On balance, there seems to be little cause for worry about weak instruments in 30-instrument models, even with the age quadratic included.

The most worrisome specifications are those reported in columns 5 and 6. These estimates were constructed by adding 150 interactions between quarter of birth and state of birth to the 30 interactions between quarter of birth and year of birth. The rationale for the inclusion of state-of-birth interactions in the instrument list is to exploit differences in compulsory schooling laws across states. But this leads to highly overidentified models with 180 (or 178) instruments, many of which are weak. The first stage F -statistics for these models are only 2.6 and 2.0. On the plus side, the LIML estimates again look fairly similar to 2SLS. Moreover, the LIML standard errors are not too far above the 2SLS standard errors in this case. This suggests that you can't always determine instrument relevance using a mechanical rule, such as " $F > 10$." In some cases, a low F may not be fatal.⁴⁴

Finally, it's worth noting that in applications with multiple endogenous variables, the conventional first-stage F is no longer appropriate. To see why, suppose there are two instruments for two endogenous variables and that the first instrument is strong and predicts both endogenous variables

⁴⁴Cruz and Moreira (2005) similarly conclude that low F -statistics notwithstanding, there is little bias in the Angrist and Krueger (1991) 180-instrument specifications.